CanSat PCB Design Guide (Team Verticality)

Rev 2 · 8 Nov 2026 · original by @Hui · rev 2 aligns the rules with **CanSat.kicad_pcb** and **CanSat.kicad_dru** after the layout audit

What changed in rev 2

- **Vias:** standard 0.6/0.3, allowed minimum 0.45/0.2 everywhere, 0.4/0.2 only in the RP2040 fanout area. JLCPCB's current via pricing treats 0.2 mm drill with 0.45 mm pad (and 0.3/0.6) as standard on 4-layer boards. Confirm on the capability page at upload.
- **HighCurrent class fixed:** it now lists the real current paths (heater drain, servo return, VBAT branches), not the gate and signal nets. Minimum width 0.6 mm (about 1.5 A on 1 oz outer copper, IPC-2221, 10 °C rise), 1.0 mm where space allows. GND vias at Q1 and Q2 are now a stated rule.
- **Relaxed to match what the hardware needs:** QSPI length match 12 mm (was 5 mm), USB length match 5 mm (was 2 mm, full-speed USB only), signal and analog minimum widths.
- **Antenna keepout rewritten** for the Ra-02 on a u.FL pigtail, with the trace-antenna case kept as an alternative.
- **Assembly flags:** full exclude list for BOM and position files, and the J2 footprint rule.
- **New:** wire pad pin-order table taken from the PCB, DRU rule corrections, and a table of known deviations in the current layout with a fix or accept decision for each.

1. Board outline and mechanical limits

The main board is a 52 mm round, 4-layer, 1.6 mm board that sits in the 22 mm top tier of the 64 mm printed shell. Sizes come from the PDR (4.1, 4.2). Items marked assumed are placeholders: take the real values from your CAD. The modules sit off-board and are wired to pads, so there are no module mounting holes. Put strain relief (hot glue or a cable tie) where each wire leaves its pad.

Item	Value	Note
Outline	Circle, diameter 52.0 mm, centred on the origin	Edge.Cuts, closed. Area about 2,120 mm² before keep-outs
Can envelope	66 mm × 115 mm	Shell 64 mm OD × 110 mm, provisional
Top tier length	22 mm, provisional	Bulkhead, board, tall parts and wire bends all have to fit
Tie-rod holes	3 × M3, 3.2 mm drill, at 120°	Layout uses radius 22.5 mm (assumed): copy the real radius from the shell CAD. The 7 mm washer keepout reaches exactly to the board edge at this radius, so any larger radius needs a smaller keepout or a smaller board
Washer keepout	7 mm diameter round each hole	Stainless rods must not touch copper
Copper to board edge	0.5 mm minimum	Parts at least 1.0 mm from the edge
Shell clearance	At least 1 mm from board edge to shell inner wall	Assumed: check the printed wall thickness
Slots near BMP280	1.0 mm wide, ends rounded, on Edge.Cuts	Layout: slot at x +16.8 to +17.8 mm, y -3.5 to +7.5 mm, between the ADXL375 and BMP280 pads. No track may cross it
Wire pads	1.8 mm pad, 1.0 mm drill	Group by destination, near the board edge. Keep every pad at least 1 mm inside the edge
Tallest part	100 µF radial (C22), 7 mm	Check the rest in the 3D viewer against the 22 mm tier

Wires run down the shell wall to the two lower tiers. **Bottom tier** pads: battery, switch, servo and the heater driver side (Q1, R8). **Middle tier** pads: NTCs and light sensors. The heater resistor R8 itself sits on the chamber block, so its two wires run up to the middle tier even though its pads are at the battery edge next to Q1. That keeps the switched current loop short. Rev 1 listed the heater in both groups; this is the single rule now.

2. Stackup, design rules and net classes

Stackup (4 layers, 1.6 mm): F.Cu signal and parts, In1.Cu solid GND, In2.Cu 3V3 pour, B.Cu signal and short return paths. VBAT stays on tracks (0.5 to 0.8 mm), not on a pour. Use the JLCPCB JLC04161H-7628 stackup (about 0.21 mm prepreg between layers 1 and 2, so the GND plane is close to the top-side traces). Enter it in Board Setup. The layout already has it.

Fab limits to set in Board Setup. Check against the JLCPCB capability page before ordering:

Rule	Value	Why
Minimum clearance	0.15 mm	Safe margin over the 0.09 mm fab minimum

Rule	Value	Why
Minimum track width	0.15 mm	Same. 0.127 mm only inside RP2040_FANOUT
Standard via	0.6 mm pad / 0.3 mm drill	Standard, no extra fee
Minimum via (anywhere)	0.45 mm pad / 0.2 mm drill	Listed as standard on JLCPCB 4-layer. Set this as the board minimum so DRC matches
Fanout via (RP2040_FANOUT and U1 exposed pad only)	0.4 mm pad / 0.2 mm drill	Needed for 0.4 mm pitch. A 0.4 pad on a 0.2 drill is not confirmed as free: check the DFM result at upload
Hole to hole	0.5 mm	Fab rule
Copper to edge	0.3 mm (fab), 0.5 mm in practice	Use 0.5 mm
Silk to pad	0.15 mm	Keeps silk off pads
Tenting	All vias tented	Avoids solder wicking on top-side-only assembly

Net classes. Net classes live in the .kicad_pro file, so create them there and use these names exactly:

Class	Nets	Width	Clearance	Notes
Default	Everything not listed, including gate and signal nets: /HEATER, /SERVO_P, /SERVO_S, Net-(Q1-G), Net-(Q2-G), I2C, LED, INT, CS	0.2 mm (0.15 min)	0.15 mm	Slow signals. The gate nets were wrongly in HighCurrent in rev 1
Power	+3V3, /VBAT, /VBAT_R, /VREG_VOUT	0.5 mm (0.3 mm where crowded, 0.15 mm only for stubs under 3 mm to a cap)	0.2 mm	3V3 on the In2 pour, VBAT on tracks
HighCurrent	Net-(Q1-D) heater drain, Net-(J6-Pin_1) servo return, the VBAT branches to R8, J6 and C22	0.6 mm min, 1.0 mm where space allows	0.25 mm	Heater about 0.28 A, servo stalls near 1 A. Keep short. GND at Q1 and Q2 is handled by vias (section 5)
USB	/D+, /D-, Net-(U1-USB_DP), Net-(U1-USB_DM)	0.2 mm	0.2 mm	Route as a pair, lengths within 5 mm (full-speed 12 Mbps), no stubs
QSPI	/SS, /SD0 to /SD3, /SCLK	0.15 mm (0.127 mm inside the fanout)	0.2 mm	Each under 20 mm, all on F.Cu, no vias. Lengths within 12 mm of each other
Crystal	/XIN, /XOUT, Net-(U1-XOUT)	0.15 mm min, 0.2 mm target	0.3 mm	Short, top layer only, guarded by GND
SPI fast	/RA_MOSI, /RA_MISO, /RA_SCLK, /IMU_MOSI, /IMU_MISO, /IMU_SCLK	0.15 mm	0.2 mm	Keep away from analog and the antenna
Analog	/BATT_SENSE, /NTC_A, /NTC_B	0.2 mm target, 0.15 mm min	0.3 mm	Away from heater and servo traces, guard with GND, stay on F.Cu where possible

Use the 0.127 mm tracks and 0.4 mm vias only in the RP2040 fanout rule area (section 3).

3. Rule areas and what each enforces

Draw each area with Place > Rule Area, name it exactly as shown, and use the settings in the table. Areas marked custom rule need an entry in the .kicad_dru file. Run DRC once on a test board, because the syntax is untested in KiCad 10.

#	Name	Covers	Type	Rules
1	RP2040_FANOUT	10 mm square round U1 (layout: x ±5.2, y -2.6 to +7.6 around U1)	Custom rule	Clearance 0.127, track 0.127, via 0.4/0.2 allowed, tented
2	RP2040_EP_VIAS	Exposed pad of U1 (pin 57)	Via array, not a keepout	3 × 3 vias, 0.2 mm drill, 1.0 mm pitch, tented, all GND. The layout has this
3	XTAL_KEEPOUT	Y1, C10, C11, R1 and 3 mm around them	Keepout: no vias, no pour on F.Cu, tracks allowed (XIN/XOUT only)	GND guard on F.Cu and solid GND on In1 below. **Also cut the In2 3V3 pour out under it**
4	VREG_LOOP	VREG_VIN and VREG_VOUT caps, U1 pins 43 to 45	Custom rule	Tracks at least 0.3 mm, cap within 3 mm of the pin, one via to GND per cap
5	QSPI_ZONE	U2 flash and U1 QSPI pins	Custom rule	No analog or high-current tracks inside. Length and layer rules are checked by hand

#	Name	Covers	Type	Rules
6	USB_ZONE	USB pads, R2, R3	Custom rule	Pair skew 5 mm max (use the skew constraint, section 3.1)
7	RF_ANTENNA_KEEPOUT	See the antenna note below	Keepout	See below
8	HEATER_SERVO_ZONE	Q1, Q2, C22, J1 side, heater and servo pads	Custom rule	HighCurrent class only, no analog, USB or crystal nets within 3 mm
9	ANALOG_QUIET	Battery divider, NTC pull-ups, ADC pins GP26 to GP28, cap near U1 pin 43	Custom rule	Analog class only, no switching nets crossing, solid GND under
10	WIRE_PADS	All THT wire pads (modules, battery, servo, heater, NTC)	Keepout: no vias	**Draw one area per pad group**, not only the NTC pads (layout has only that one). Strain relief space, no small parts
11	MECH_KEEPOUT	3 × M3 rod holes, 7 mm diameter	Keepout: no copper, parts, tracks or vias	Allows only the drill hole (H1 to H3)
12	BMP_THERMAL_ISO	Slots and 5 mm round the BMP280 wire pads	Keepout: no copper pours on any layer	Keeps board copper from conducting heat into the sensor
13	ASSEMBLY_EDGE	1.0 mm band inside the outline	Keepout: no parts	Round board with no rails: ask JLCPCB to confirm their process handles it

Antenna keepout (rev 2). The BOM uses the Ra-02 with a u.FL to bare-wire pigtail. In that case the antenna is off-board, so keep parts and tracks at least 5 mm away from the module's u.FL and antenna end, keep the pigtail routed away from the board and heater, and the solid In1 GND may stay under the module. If you ever use the module's on-board trace antenna (Ra-01 style), switch to the original rule: no copper on any layer under the antenna end and nothing within 5 mm. The layout currently has a thin strip at y -24.8 to -23.1 only, so redraw it for whichever case you pick.

3.1 Custom rules for the .kicad_dru file

```
(version 1)
(rule "RP2040 fanout"
  (condition "A.insideArea('RP2040_FANOUT')")
  (constraint clearance (min 0.127mm))
  (constraint track_width (min 0.127mm))
  (constraint via_diameter (min 0.4mm))
  (constraint hole_size (min 0.2mm)))

(rule "Via size outside fanout"
  (condition "A.Type == 'Via' && !A.insideArea('RP2040_FANOUT')")
  (constraint via_diameter (min 0.45mm))
  (constraint hole_size (min 0.2mm)))

(rule "HighCurrent width"
  (condition "A.NetClass == 'HighCurrent' && A.Type == 'Track'")
  (constraint track_width (min 0.6mm)))

(rule "USB skew"
  (condition "A.NetClass == 'USB'")
  (constraint skew (max 5mm)))

(rule "No pour at BMP280"
  (condition "A.insideArea('BMP_THERMAL_ISO') && A.Type == 'Zone'")
  (constraint disallow zone))
```

The old "USB pair skew" rule used diff_pair_uncoupled, which measures uncoupled length, not length difference. Replace it with skew as above. Keep the existing rules in CanSat.kicad_dru for clearance, VREG_LOOP, QSPI_ZONE, analog quiet and MECH_KEEPOUT. Where an area only needs a pure keepout (7, 10, 11, 12, 13), tick the keepout options in the Rule Area dialog instead of writing a custom rule.

4. Placement rules and order

Place in this order, locking each group before starting the next. All SMT parts go on F.Cu only, because JLCPCB Economic assembly is top-side only.

Assembly flags (rev 2). Tick Exclude from BOM and Exclude from position files on every part JLCPCB does not place, and check the flags are on the PCB footprints, not just the schematic symbols (run Update PCB from Schematic):

- Hand-fit: U3, U5 (Ra-02), C22, J1, SW1, R8
- Modules and wire pads: U4, U6, U7, U8, J2 to J6, TH1, TH2, TP1 to TP5, H1 to H3

- Do not place: R18 to R21 (4.7 k, DNP) - exclude from both as well as DNP
- J2 must use the CanSat_Modules:Pad_2pad_GND_SIG footprint (pad only, no LCSC part). It is currently an 0402 resistor footprint
- The remaining 157 SMT joints per board are placed by JLCPCB

Order

- 1. Outline, M3 holes and slots (section 1).
- 2. U1 RP2040 near the centre, offset slightly towards the USB and SWD pads.
- 3. U2 flash within 10 mm of U1, on the QSPI side.
- 4. Y1, C10, C11, R1 within 5 mm of XIN and XOUT.
- 5. Decoupling caps on U1 (table below).
- 6. Power: U3 MCP1700 pads, input and output caps, near the battery pads.
- 7. Ra-02 (U5) near the outer edge on the antenna side.
- 8. Heater and servo drivers (Q1, Q2, C22) on the battery side, far from U1 analog pins.
- 9. Analog divider and NTC pull-ups beside the ADC pins.
- 10. Wire pads grouped at the edge by destination.
- 11. LED, R17, test points, silk last.

Zones on the 52 mm disc

Zone	Contents	Rule
Centre	U1, U2, crystal	Keep more than 8 mm from the heater and servo drivers (measured from the U1 centre)
Antenna edge	U5 Ra-02, u.FL, antenna wire	See the antenna note in section 3. Keep other parts and test points out of the 5 mm zone where the layout allows. In the current layout TP4, TP5, J3, C20 and TH2 are inside it
Battery edge	J1, SW1, U3, C22, Q1, Q2, R8 and servo pads	All high-current parts together, shortest loop to the battery pads
Sensor edge	BMP280 and ADXL375 wire pads, slots	Away from heat sources, 20 mm or more from the battery edge where the board allows (layout: 17 to 19 mm, accepted, the heater itself is on the chamber block)
Analog corner	R divider, NTC resistors, ADC caps	Between U1 and the sensor edge, away from the battery edge
Middle-tier edge	NTC and TSL pads	Together so the wires run as one bundle

Part	Rule
100 nF on each IOVDD pin (1, 10, 22, 33, 42, 49)	Within 2 mm of the pin, via to GND right at the cap pad
DVDD caps (pins 23 and 50)	Within 3 mm, short to GND
VREG_VIN and VREG_VOUT caps (pins 44, 45)	Within 3 mm, 1 µF on each
USB_VDD cap (pin 48)	Within 3 mm
ADC_AVDD cap (pin 43)	Within 3 mm, ties to quiet 3V3 not the noisy rail
C22 (100 µF)	Beside Q2, in the servo supply path
R2, R3 (22 ohm USB series)	Within 5 mm of U1 pins 46 and 47, not at the pads
R17 (servo pull-down)	Beside the servo wire pad
Crystal	One layer only, no vias in its connection, no traces underneath
Pull-up and series resistors	At the receiving end of the signal

Orient all two-pin parts in the same direction. Keep anything taller than 5 mm (C22) off the antenna edge and inside the 22 mm tier height budget. Leave 0.5 mm between parts, and 1 mm for parts fitted by hand. Put polarity and pin 1 marks on the silkscreen for U3, SW1, J1 and C22.

Wire pad pin order (from CanSat.kicad_pcb)

U4, U6, U7 and U8 use chip-pin footprints, so solder each wire to the module pin that carries the same signal, not to the same pin number.

Ref	Pad: signal
U4 ADXL375	1 3V3, 2 GND, 4 GND, 5 GND, 6 3V3, 7 IMU_CS, 8 IMU_INT1, 12 IMU_MISO (SDO), 13 IMU_MOSI (SDI), 14 IMU_SCLK. Pads 3, 9, 10, 11 not connected
U6 BMP280 (SPI)	1 GND, 2 PRESS_CS (CSB), 3 IMU_MOSI (SDI), 4 IMU_SCLK (SCK), 5 IMU_MISO (SDO), 6 3V3 (VDDIO), 7 GND, 8 3V3 (VDD). Needs an SPI-capable module (6-pin GY-BMP280, not the 4-pin I2C-only type)
U7 TSL2591 A	1 SCL0, 2 INT_A, 3 GND, 5 3V3, 6 SDA0. Pad 4 not connected
U8 TSL2591 B	1 SCL1, 2 INT_B, 3 GND, 5 3V3, 6 SDA1. Pad 4 not connected

Ref	Pad: signal
J1 battery	1 VBAT_R (battery +), 2 GND. Check against the LiPo plug
SW1	1 VBAT, 2 VBAT_R (common), 3 not connected
J3 SWD	1 SWDIO, 2 GND, 3 SWCLK
J4 USB	1 GND, 2 D-, 3 D+. Never the red 5 V wire
J5 RUN	1 GND, 2 RUN
J6 servo	1 GND (Q2-switched), 2 VBAT, 3 SIG (SERVO_S)
R8 heater	1 Q1 drain, 2 VBAT
TH1, TH2	1 ADC node (NTC_A, NTC_B), 2 GND

5. Routing rules by net group

Group	Rule
Ground	Solid In1.Cu plane, no splits, no slots under U1, the crystal or the Ra-02. Stitch GND vias every 5 to 8 mm and around the board edge. Every GND pad gets its own via, not a shared long trace
Power	Feed from the plane. 3V3 pour on In2, cut out under the crystal and the antenna end. VBAT on tracks. 3V3 enters U1 from a short top-layer trace, one via per supply cap
RP2040 fanout	Dog-bone vias outwards from the pads, 0.127 mm tracks inside the rule area only, escape the 4 sides to separate layers
Crystal	Top layer only. XIN and XOUT as short as possible, equal length within 2 mm, no vias, GND guard around, nothing routed underneath on F.Cu. 1 k series resistor on XOUT right at the pin
QSPI	Every signal on F.Cu, no vias, each under 20 mm, lengths within 12 mm of each other (the RP2040 boots the flash at a low clock). Series resistor at the RP2040 end if the clock rings
USB	Route D+ and D- as a pair from U1 through R2, R3 to the pads. Lengths within 5 mm, no stubs, as few vias as possible, GND plane unbroken below
SPI0 (Ra-02)	MOSI, MISO, SCK together, short, away from the antenna side, GND between them and any analog net
SPI1 (ADXL375 and BMP280)	Shared bus, stubs under 5 mm to the wire pads. Drive both CS lines high at boot (firmware) before the bus is used
I2C (TSL2591 A and B)	Two separate buses, pull-ups within 5 mm of the RP2040 or at the wire pad. Keep clear of the heater
Analog (ADC)	Divider and NTC lines short, over solid GND, no digital lines parallel within 1 mm, guard with GND
Heater and servo	HighCurrent class: 0.6 mm minimum, 1.0 mm where space allows, shortest return path. **At least three 0.6/0.3 GND vias on each of the Q1 and Q2 source pads** (a single 0.2 mm via is not enough for the servo stall current). Keep away from the BMP280 and analog nets. The FET body diode covers servo switch-off spikes, add a flyback diode if the servo is changed
Antenna	Keep the pigtail short and away from the heater and servo wires. Nothing routed under or beside the antenna end
Test and SWD	SWDIO and SWCLK as short top-layer stubs to the pads, off the USB pair

Route at 45° or arcs, never 90° turns on fast nets. Change layers with a via next to a GND via where a signal crosses a plane change. Prefer F.Cu for signals and keep B.Cu for short links. Keep traces at least 0.5 mm from the board edge. Fill zones last, then run DRC and look at the planes in the 3D viewer for islands. **Add teardrops** on the 0.127 mm fanout pads and on the wire pads (they take mechanical stress). The current layout has none.

6. Fabrication and assembly outputs (JLCPCB Economic)

Order settings: 4 layers, 1.6 mm, 52 mm round outline (Edge.Cuts closed, no gaps), quantity 5 boards minimum, assembly on the top side only, 2 boards assembled, tented vias, ENIG (as in the layout) or HASL if the budget needs it. ENIG is flatter for the RP2040 pad, HASL is cheaper.

File	How	Notes
Gerbers	Plot all copper layers, F.Mask, B.Mask, F.Paste, F.Silkscreen, B.Silkscreen, Edge.Cuts	Protel names, "Plot footprint values" off, tick "Use drill/place file origin" (the aux origin is the board centre)
Drill files	Excellon, merged PTH and NPTH	Same origin as the Gerbers
BOM	CSV with Designator, Comment, Footprint, LCSC Part #	Only parts JLCPCB places. Parts flagged Exclude from BOM are dropped
Pick and place	Position file, CSV, top side only, mm	Parts flagged Exclude from position files are dropped. Use the same origin as the Gerbers

LCSC part numbers to enter in the schematic (placed parts only): U1 C2040 (Extended), U2 C97521, Y1 C9002 (rename the value from Crystal_GND12 to 12MHz), C10 and C11 C1562, 100 nF C1525, 1 µF C52923, 10 µF C19702, 1 k C11702, 22 ohm C25092, 10 k C25744, 100 ohm C25076, 100 k C25741, AO3400A C20917, LED C2297.

Checks after upload: in the JLCPCB 3D preview, check the rotation of U1, U2, the LED and the transistors (KiCad and JLCPCB conventions differ), the crystal orientation and the AO3400A pin 1 position, and that nothing is placed in the antenna zone. Check the DFM result for the 0.4/0.2 fanout vias. Ask JLCPCB to confirm that a round outline with no rails is accepted for assembly, since they may add rails or a panel at extra cost. The 52 mm disc is well inside the under 100 × 100 mm price band.

7. Pre-order DRC and review checklist

Schematic

- Chip symbols for U4, U6, U7, U8 point to the *_chippins footprints, with no pad-not-found errors on netlist update.
- R2 and R3 are 22 ohm. NC flags on U7 and U8 pin 4. ERC clean apart from known notes.
- Every placed part has an LCSC field.
- GP0 and GP1 UART labels (RX and TX) checked against the MCU direction.
- BMP280 module is SPI-capable and PRESS_CS goes to U1 pin 14.

PCB

- Edge.Cuts is one closed 52 mm circle with M3 holes and the slot.
- Net classes (with the corrected HighCurrent list) and the 13 rule areas are drawn and named exactly.
- Board minimum via is 0.45/0.2 and the only smaller vias are inside RP2040_FANOUT.
- DRC has zero errors, and any warnings are understood and noted.
- GND and 3V3 planes have no islands, and zones were refilled before DRC.
- 3V3 and VBAT cannot short at the wire pads.
- Crystal, USB and QSPI routed per section 5, with lengths checked.
- Q1 and Q2 each have three 0.6/0.3 GND vias. Heater drain and servo return are 0.6 mm or wider.
- In2 3V3 pour is cut out under the crystal. Antenna keepout redrawn for the chosen antenna.
- Teardrops added. 3D viewer: tallest part fits the 22 mm tier, board fits the shell with at least 1 mm.

Outputs and ordering

- Exclude flags set on the PCB footprints (section 4 list) and J2 uses the pad-only footprint.
- Gerbers, drills, BOM and position files regenerated after the last change.
- JLCPCB preview shows every placed part with the right rotation and no missing parts.
- Cost matches the plan (total about £154, one Extended part fee).

Bring-up order after the boards arrive

- 1. Check 3V3 and GND with a meter for shorts before connecting a battery.
- 2. Power from USB or a bench supply limited to 100 mA, and check 3V3 is 3.3 V.
- 3. Flash over USB (hold BOOTSEL), then blink the LED.
- 4. Test the crystal by reading a timer against a known time.
- 5. Add modules one at a time: BMP280, ADXL375, TSL2591, NTCs, Ra-02.
- 6. Last, connect heater and servo, and check the servo ground path with a scope or meter.

Appendix: known deviations in the current layout (audit, 8 Nov 2026)

Item	Layout now	Decision
Assembly flags	Only H1 to H3 flagged. Others missing or BOM-only	**Fix** (section 4)
J2 footprint	0402 resistor	**Fix**: Pad_2pad_GND_SIG
Q2 GND via	1 via, 0.45/0.2	**Fix**: three 0.6/0.3
Q1 GND vias	2 vias, 0.4/0.2	**Fix**: three 0.6/0.3
Heater drain and servo return width	0.5 to 0.6 mm	Accept at 0.6 mm, widen where space allows
Vias outside fanout	58 vias at 0.4/0.2	**Fix**: grow to 0.45/0.2 or 0.6/0.3

Item	Layout now	Decision
Antenna keepout	Thin strip at the edge only	**Decide** antenna type, then redraw (section 3)
Test points and SWD near the Ra-02	TP4, TP5, J3, C20, TH2 within 5 mm	Accept if the pigtail antenna is used
QSPI length spread	SD0 19.0 mm, SD2 7.6 mm	Accept (rule relaxed to 12 mm)
USB length	D+ about 17.6 mm, D- about 13.6 mm, uses B.Cu and vias	Accept (rule relaxed to 5 mm), reduce vias if easy
Teardrops	None	**Fix**
In2 3V3 pour	Full disc, under the crystal	**Fix**: cut out under Y1, C10, C11
Analog trace width	0.15 mm, NTC lines use B.Cu and vias	Accept 0.15 mm min, keep guarded
Y1 value	Crystal_GND12	**Fix**: 12MHz
WIRE_PADS area	Covers the NTC pads only	**Fix**: one area per pad group